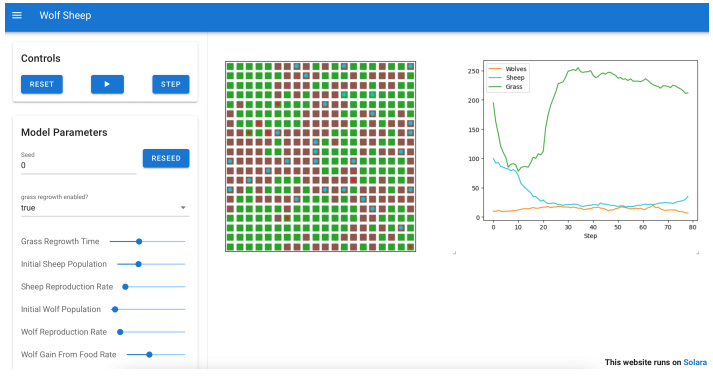
Foundations of Agent-based systems
Session 5 – Project

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Agenda

1. **2026/09/07** – Introduction and basics
   - 1:30 lecture: basics of MAS
   - 0:30 tutorial: MAS modeling
   - 1:00 practical session: MAS in Java
2. **2026/09/14** – Communication and platforms
   - 1:00 lecture: communication, platforms
   - 1:00 practical session: MAS in Java (cont.)
   - 1:00 tutorial : JADE
3. **2026/09/28** – Practical session : JADE
4. **2026/10/05** – MultiAgent Based Simulation
   - 1:00 lecture: theory of MABS
   - 2:00 practical session: MESA

➜ **2026/10/12** – Practical session : MABS project

6. **2026/10/19** – Agentic AI (practical session)

Session 5
MABS in Mesa or JADE

Project description

Goal

MultiAgent Based Simulation to study a concrete problem

Organisation

- ✓ Groups of 1 or 2 students
- ✓ 6h work per student (3h today, 3h homework)
- ✓ Development: Python + mesa or Java + Jade
  - ✗ Other MAS platforms (Repast, Gama...) not allowed
  - ✗ Home-made MAS without library not allowed

Deadline

Sunday, October the 25th, 23:59 CET

Project description

Goal

MultiAgent Based Simulation to study a concrete problem

Expectations

- Source code
- 5-10 pages max report

Evaluation (0-20 grade)

- Design choices 20% (to be explained in the report)
- Implementation 20% (source code & correct use of the library)
- Features & useability 20% (I will run the programs...)
- Analysis 20% (simulation results in the report)
- Overall presentation 20% (report's organization, comments in code, etc)

Project description

Mission

In order to reduce pollution, the city council wants to test a new carshare solution for taxi. Your role is to evaluate the possible impact of such a solution on the trafic, passenger's travel time and acceptability.

Methodology

- Design should be incremental, e.g.
  1. single-passenger transport
  2. two-passengers with no route change
  3. two-passengers with route change & multi-hop
  4. multi-passengers, multi-hop, etc
- Identify simulation parameters **early**
- What is the question (observable variables)?
- Make it simple! (and even more simpler)

Project description

Questions

- How to model car that transport passengers in the city, without going into too many details?
  - E.g. multigrid with each cell being one district of the city
- How to model people's movements in the city
  - E.g. can people walk from a district to the next one?
- How to model taxishare and each passenger's route?
- How to model delais when reaching a new passenger?
- How to model simulate travel time?
- How to study acceptability (opt.)?
- How to model possible trafic jam (opt.)?

Project in Mesa

Mesa: some recommendations

✓ Use `MultiGrid`

You want to have multiple cars/passengers/buildings on each cell

✗ Do not necessarily use the grid as a map!

- A district can be represented by one or by multiple cells, some cells can represent connections or highways
- Vehicules can jump over cells (otherwise you need a routing algorithm)
- Time to jump (internal step counter) = *e.g.* car speed or district size or distance...

✓ Minimal direct communication is required

Passengers need to call for a taxi: CFP + answers

✓ Use visualization interface from the begining

View positions, draw graphs...

Project in Jade

Jade: some recommendations

- ✗ Do not implement a user interface
  Logs variable values in a file and use a spreadsheet
- ✓ In JADE, everything is an agent!
  Use "district" agents (and possibly connection agents, highway agents...)
- ✓ Use timers to for speed, trafic, boarding delays...
  WakerBehaviour is your friend
- ✓ Verify CPU usage!

Let's begin!